

Research Interests

Madden–Julian Oscillation (MJO); tropical waves; subseasonal-to-seasonal (S2S) prediction; tropical cyclone genesis; climate extreme; climate variability and climate change.

Education

2023–Present **Ph.D. in Atmospheric Science, Colorado State University, U.S.**

Advisor: Eric D. Maloney GPA: 3.9/4.0

- Focusing on dynamics and predictability of East Pacific easterly waves, including their interactions with the MJO, moisture–vorticity coupling, and tropical cyclone genesis in S2S forecast systems.

Sep 2019–Jun 2021 **M.S. in Atmospheric Science, National Central University, Taiwan**

Advisor: Jia-Yuh Yu GPA: 3.9/4.0

Sep 2015–Jun 2019 **B.S. in Atmospheric Science, National Central University, Taiwan**

Research Appointments

Apr 2026–May 2026 **Research Visitor, Australian Bureau of Meteorology, Melbourne, Australia**

Host: Matthew Wheeler

- Collaborated with ACCESS-S2 development and research scientists to evaluate tropical wave prediction and sources of forecast skill in subseasonal prediction systems.

Publications

First-Author Publications

- [1] Tsai, Y.-C. and E. D. Maloney, 2026: Representation of East Pacific Easterly Waves in Subseasonal-to-Seasonal Forecasts of the Unified Forecast System. *J. Adv. Modeling Earth Sys.*
- [2] Tsai, Y.-C., E. D. Maloney, D. Kim, and S. J. Camargo, 2025: Unified Forecast System Model Prediction of the Madden–Julian Oscillation and Its Influence on East Pacific During Boreal Summer. *J. Geophys. Res.*, 130, e2024JD042904.
- [3] Tsai, Y.-C., Li, J.-L., K.-M. Xu, W.-L. Lee, J. H. Jiang, E. J. Fetzer, and J.-Y. Yu, 2023: Possible linkage of sea surface height anomaly, surface wind stress and sea surface temperature with the falling ice radiative effects under a gradual warming scenario. *Environ. Res. Commun.*, 5(8), 085004
- [4] Tsai, Y.-C. and J.-Y. Yu, 2023: Contrasting the Energy Recharge–Discharge Cycle between Propagating and Eastward-decaying Madden–Julian Oscillation Events. *Climate Dynamics*, 61, 2565–2579.

Co-Author Publications

- [5] Li, J.-L., Lee, W.-L., Xu, K.-M., Jiang, J.H., Wang, Y.-H., Y.-C. Tsai, Ho, S.-P., et al., 2026: Attributing upper-tropospheric warm biases in CMIP6 models to ice cloud-radiation interaction deficiencies over tropical oceans. *Geophys. Res. Lett.*, 53(9), e2025GL120130.
- [6] Chang, C. J., M. Lo, W. Tseng, Y.-C. Tsai, and J. Yu, 2024: Impact of Deforestation in the Maritime Continent on the Madden–Julian Oscillation. *J. of Climate*, 37, 2247–2261.
- [7] Li, J.L.F., Wang, L.C., Tsai, Y.C., Huang, Y.S., Lee, W.L., Jiang, J.H., et al., 2024: Exploring the relationship between upper ocean states and the falling Ice radiative effects using ECCO product and global climate models. *Environ. Res. Commun.*, 6(12), 121009.
- [8] Li, J.-L. F., Xu, K.-M., Y.-C. Tsai, Lee, W.-L., Jiang, J. H., et al., 2023: Evaluation of radiatively active frozen hydrometeors mass in CMIP6 global climate models using CloudSat-CALIPSO observations. *J. Geophys. Res.*, 128, e2023JD039200.

- [9] Li, J.-L. F., Xu, K.-M., Lee, W.-L., Jiang, J. H., **Y.-C. Tsai**, Yu, J.-Y., et al., 2023: Warm clouds biases in CMIP6 models linked to indirect effects of falling ice-radiation interactions over the tropical and subtropical Pacific. *Geophys. Res. Lett.*, 50, e2023GL104990.
- [10] Li, J.-L., W.-L. Lee, K.-M. Xu, **Y.-C. Tsai**, J. H. Jiang, J.-Y. Yu, et al., 2023: Radiatively Active Hydrometeors Frequencies from CloudSat-CALIPSO Data for Evaluating Cloud Fraction in Global Climate Models. *J. Geophys. Res.*, 128, e2023JD038511.
- [11] Li, J.-L., **Y.-C. Tsai**, K.-M. Xu, W.-L. Lee, J. H. Jiang, J.-Y. Yu, et al., 2022: Inferring the Linkage of Sea Surface Height Anomalies, Surface Wind Stress and Sea Surface Temperature with the Falling Ice Radiative Effects Using Satellite Data and Global Climate Models. *Environ. Res. Commun.*, 4(12), 125004.
- [12] Li, J.-L., K.-M. Xu, W.-L. Lee, J. H. Jiang, **Y.-C. Tsai**, E. J. Fetzer, et al., 2022: Comparing surface wind stress and sea surface temperature biases over the tropical and subtropical oceans in subsets of CMIP6 models categorized by frozen hydrometeors-radiation interactions. *Environ. Res. Commun.*, 4(5), 055009.

Selected Presentations

- 2026 **Oral:** Representation of East Pacific Easterly Waves in S2S Forecasts of the Unified Forecast System. AMS Tropical.
- 2026 **Invited Talk:** Seminar at Bureau of Meteorology, Melbourne, Australia.
- 2026 **Invited Talk:** TROPICOOOL Seminar, University of Wisconsin–Madison, USA.
- 2026 **Invited Talk:** Seminar at Research Center for Environmental Changes & NCU, Taiwan.
- 2025 **Poster:** Unified Forecast System Model Prediction of the MJO and East Pacific Teleconnections During Boreal Summer. American Meteorological Society Annual Meeting.
- 2023 **Poster:** Contrasting the Energy Recharge–Discharge Cycle between Propagating and Eastward-decaying MJO Events. Asia Oceania Geosciences Society Annual Meeting.

Honors and Awards

- 2026 ICTP Travel Grant Recipient, 2nd Workshop on Organisation and Maintenance of Tropical Convection and the MJO, Trieste, Italy.
- 2026 UIFCW Student Travel Grant, Unifying Innovations in Forecasting Capabilities Workshop.
- 2025 William Gray Tropical Meteorology Award, Colorado State University.
- 2025 ATS ASCENT International Research Experience Scholarship, Colorado State University.
- 2025 Student Travel Grant Recipient, Dynamical Core Model Intercomparison Project Summer School.
- 2024 UIFCW Student Travel Grant, Unifying Innovations in Forecasting Capabilities Workshop.
- 2023 2023–2025 Ministry of Education Study Abroad Scholarship, Taiwan.
- 2022 DR. CHIA CHOU Thesis Diploma Award, Taiwan.
- 2021 First Prize, Atmospheric Science Postgraduate Student Conference, Taiwan.
- 2021 Representative of the Graduates, Department of Atmospheric Science, National Central University.

Relevant Experience

- Aug 2021–Jul 2023 **Research Assistant**, *National Central University*, Taiwan
 - Conducted climate dynamics research related to the MJO, climate variability, and model evaluation.
- Aug 2019–Jul 2021 **Teaching Assistant**, *National Central University*, Taiwan
 - Lecturer of NCL tutorials in the Climate Research Project course.
- Jun 2018–Aug 2018 **Summer Intern**, *Central Weather Bureau*, Taipei City, Taiwan
 - Evaluated regional wind-speed forecasts from numerical weather prediction models.

Technical Skills

Programming Python, NCL, Fortran, Bash, LaTeX.